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REVIEW



New investigational drugs for steroid-refractory acute graft-versus-host disease: a review of the literature

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ABSTRACT

Introduction: Steroid-refractory acute graft-versus-host disease (SR-aGVHD) remains a formidable obstacle in the field of allogeneic hematopoietic cell transplantation (allo-HCT), significantly contributing to patient morbidity and mortality. The current therapeutic landscape for SR-aGVHD is limited, often yielding suboptimal results, thereby emphasizing the urgent need for innovative and effective treatments.

Areas Covered: In light of the pivotal REACH2 trial, ruxolitinib phosphate, a Janus kinase inhibitor, has gained prominence as the standard treatment for SR-aGVHD. Nevertheless, a considerable number of patients either do not respond to or cannot tolerate this therapy. This review delves into emerging treatments for SR-aGVHD, including mesenchymal stromal cells (MSCs), fecal microbiota transplantation (FMT), CD3/CD7 blockade, neihulizumab, begelomab, tocilizumab, and vedolizumab. While some of these agents have shown encouraging results in early-phase trials, issues such as treatment-related toxicities and inconsistent responses in larger studies highlight the necessity for ongoing research.

Expert Opinion: Current trials exploring new agents and combination therapies offer hope for fulfilling the unmet clinical needs in SR-aGVHD, potentially leading to more effective and precise treatment strategies.

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GvHD; steroid refractory; ruxolitinib; mesenchymal stem cells; tocilizumab

1. Introduction

Allogeneic hematopoietic cell transplantation (allo-HCT) stands as a potentially curative treatment option for a diverse range of conditions, including hematologic malignancies, nonmalignant hematologic disorders, metabolic storage diseases, and immune deficiencies. In addition to advances in conditioning strategies and antimicrobial prophylaxis, the widespread adoption of post-transplant cyclophosphamide (PTCy) for prophylaxis of graft-versus-host disease (GVHD) has allowed for a reduction in high grade acute GVHD and an expanded use of HLA-haploidentical HCT [1].

Despite advances in prophylactic approaches [1,2], acute graft-versus-host disease (aGVHD) remains a major contributor to non-relapse mortality (NRM), affecting 30% to 50% of all patients who receive allo-HCT [3]. Previously, aGVHD has been classically defined as symptom onset occurring within 100 days following allo-HCT, whereas chronic GVHD (cGVHD) is characterized by symptom onset beyond 100 days after allo-HCT. The differentiation between acute and chronic GVHD is now rooted in the distinctive pathophysiology and clinical manifestations of the two conditions [4].

Acute GVHD is believed to stem from treatment-induced tissue injury and the subsequent release of proinflammatory cytokines, triggering the activation of donor T cells targeting recipient major and minor histocompatibility antigens. After activation, these donor T cells migrate to target organs and recruit additional effector cells, including cytotoxic T cells and

natural killer (NK) cells. These effector cells induce further tissue damage both through direct cytotoxic effects and through fueling the inflammatory cascade with cytokine release [3]. The most frequently affected organs are the skin, the gastrointestinal tract, and the liver [3]. Severity of the disease at onset and the extent of organ involvement serve as criteria to categorize aGVHD into four subtypes (mild (I) to very severe (IV)), which are indicative of both prognosis and response to treatment.

Systemic corticosteroids are the primary treatment for aGVHD; however, around 40% of patients encounter the conundrum of steroid-refractory aGVHD (SR-aGVHD) [5]. As outlined by the consensus criteria established by the European Society for Blood and Marrow Transplantation, the National Institutes of Health, and the Center for International Blood and Marrow Transplant Research (EBMT–NIH–CIBMTR) Task Force, aGVHD is deemed steroid refractory or resistant if it meets one of the following conditions: (@) progression of organ involvement within 3–5 days of initiating therapy with ≥ 2 mg/kg/day of prednisone equivalent, (@) absence of improvement within 5–7 days after treatment initiation, or (@) incomplete response after more than 28 days of immunosuppressive therapy, including corticosteroids [6]. Conversely, steroid dependence refers to the failure to taper prednisone below 2 mg/kg/day following an initially successful treatment or the recurrence of aGVHD during steroid tapering.

SR-aGVHD has been classically associated with high mortality rates. In a combined analysis of 25 studies, the weighted

Article highlights

- Steroid refractory (SR)-aGVHD remains a formidable complication post-HCT and the development of innovative therapeutic approaches remains a clinical priority.
- Ruxolitinib phosphate was granted approval by the US FDA for treating SR-aGVHD in 2019 and represents a major advance.
- Biomarkers (ST2 and REG3a) can predict outcomes of steroid-resistant GVHD patients, and in the future may help develop risk adapted novel approaches for management of this complication.

average 6-month survival estimate was 49% [5]. Patients with SR-aGVHD also had elevated NRM at 2 years (65% vs 35%) compared to patients with steroid responsive disease [7]. Therefore, developing effective therapy for SR-aGVHD is of the utmost clinical importance.

2. Current treatment landscape for SR-aGVHD

Ruxolitinib phosphate was granted approval by the US FDA for treating SR-aGVHD in 2019 [8]. Before this approval, there was a paucity of agents interrogated in phase II/III trials for SR-aGVHD, though several showed efficacies in smaller studies and are used in clinical practice.

The mainstay of SR-aGVHD treatment has historically been resolved around immunosuppressive agents inhibiting T-cell activation, such as sirolimus [9], anti-thymocyte globulin [10],

and mycophenolate mofetil [11] (Figure 1). Off-label use of monoclonal antibodies targeting cytokines has been utilized based on small prospective trials or retrospective analyses, from etanercept (anti-TNF- α antibody), to inolimomab (anti-IL2 receptor antibody). Unfortunately, these agents have failed to demonstrate a failure-free survival benefit in larger randomized trials [12].

Extracorporeal photopheresis (ECP) is a non-pharmacologic intervention which uses leukapheresis, photoactivation with ultraviolet-A (UVA), and reinfusion to activate immature dendritic cells and establish immune tolerance [13]. ECP is an established second-line therapy for cGVHD, however this intervention has demonstrated both efficacy and safety in aGVHD. In a prospective, multicenter study of ECP in children with SR-aGVHD, 16 of the 29 patients (55%) had a CR or PR at day 28 [14]. Unfortunately, there is significant variability in reported response rates for adults with SR-aGVHD, ranging from 52% to 100%, and in particular, responses reported for the liver and GI tract [15]. A meta-analysis concluded that there is insufficient evidence to recommend ECP for grade III or higher aGVHD [16]. Long-term survival based on these therapies remains limited, in part due to lack of durable response or significant infectious complications.

Ruxolitinib phosphate is an oral inhibitor targeting Janus kinases JAK1, JAK2, and tyrosine kinase 2 (Tyk2). Janus kinases are crucial in proinflammatory cytokine and growth factor receptor signaling, which are implicated in the pathogenesis of aGVHD [17]. Murine models of Interferon- γ (IFN γ) receptor –

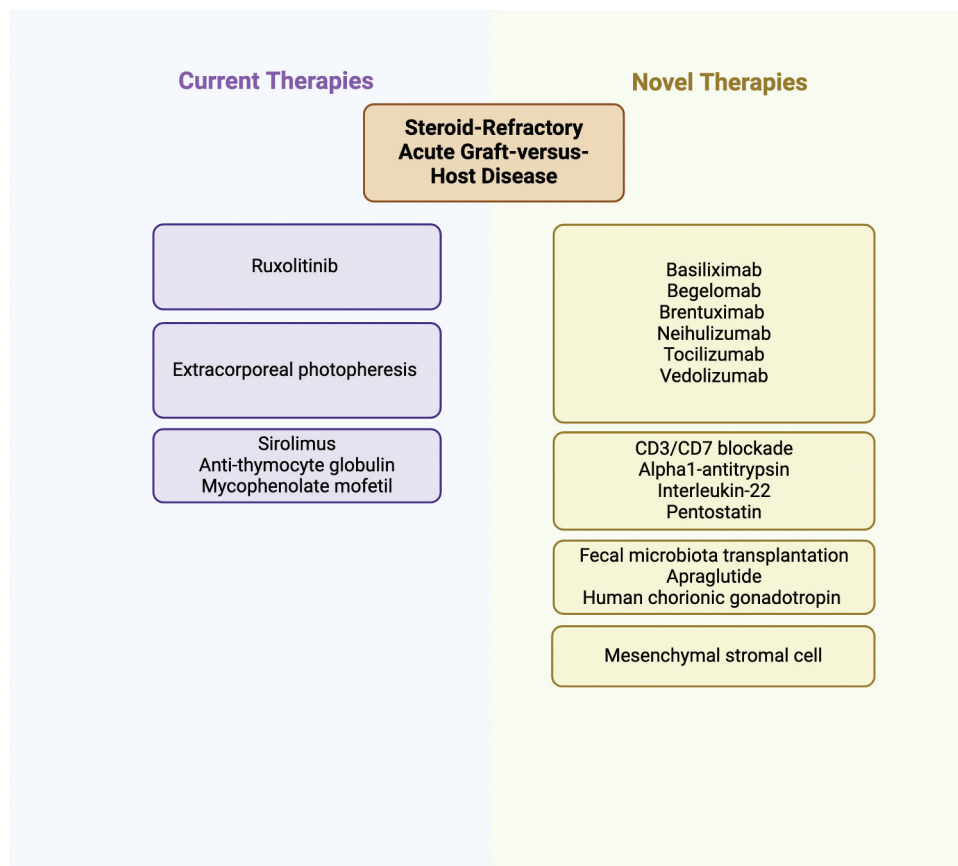


Figure 1. Current and novel therapies for SR-aGVHD.

Table 1. Ruxolitinib in SR-aGVHD.

		ORR	CR rate	Median OS (months)
Zeiser et al., 2015 [20]	Retrospective	Best: 81.5%	46%	
Jagasia et al., 2020 (REACH1) [21]	Phase II	Day 28: 54.9%	27%	7.6
Zeiser et al. 2020 (REACH2) [22]	Phase III	Ruxolitinib	34%	11.1
		Day 28: 62%		
		BAT	19%	6.5
		Day 28: 39%		
Murray et al., 2022 [24]	Retrospective	Day 28: 61%	14%	5.3

BAT: Best Available Therapy (Sirolimus, MMF, MSC, MTX, Infliximab, ECP, Etanercept, ATG); ORR: Overall response rate; CR: Complete response; OS: overall survival.

deficient allogeneic donor T cells have demonstrated a significantly lower incidence of GVHD compared to normal T cells, whilst maintaining a graft-versus-leukemia effect. Since JAK1 and JAK2 are regulators of IFN γ receptor signaling, pharmacologic inhibition with ruxolitinib was explored and resulted in an effect similar to that of interferon- γ receptor – deficient T cells [18].

Ruxolitinib emerged as a promising SR-aGVHD treatment option from a retrospective series of 54 patients with SR-aGVHD. Of those who received ruxolitinib treatment, the best ORR was 81.5%, with 46% achieving a complete therapeutic response. The 6-month survival rate of 79% observed among these patients was also noteworthy [19].

These encouraging results spurred the inception of the REACH trials to formally interrogate the efficacy and safety of ruxolitinib. REACH1, a single-arm phase II study examining 71 patients with SR-aGVHD, demonstrated a day 28 ORR of 54.9% in patients who received ruxolitinib, with CR observed in 26.8% of cases. These responses were sustained beyond the 28-day mark; 73% of patients attained a response to ruxolitinib at any time during 6 months of follow-up, with 56.3% achieving CR. The 6-month survival rate was 51% for all patients who received ruxolitinib, and 70% among treatment responders [20].

The randomized phase III REACH2 study showed the superiority of ruxolitinib over the best available therapy (BAT) [21]. The trial met its primary endpoint, demonstrating significantly improved overall response rates (ORR) at day 28 with ruxolitinib (62.3%) compared to BAT (39.4%). The median duration of response in following ruxolitinib vs. BAT was 163 and 101 days, respectively. Subgroup analysis revealed improved responses across all organ sites in patients with Grade 3 or 4 aGVHD. Ruxolitinib displayed a comparable safety profile to the BAT arm; severe adverse events were reported in 38% of patients receiving ruxolitinib compared to 34% control. Thrombocytopenia and anemia remained the most prevalent severe adverse events, while neutropenia occurred at a lower rate. The incidence of infections was nearly equivalent between ruxolitinib-treated patients and those in the control group (grade 3 or higher: 22% vs. 19%). At one year, estimated survival rates were 49% for patients on ruxolitinib and 44% for those receiving BAT. Furthermore, the success of REACH2 has led to a phase III open-label, randomized trial investigating the efficacy of ruxolitinib for cGVHD, with an improved overall response rate in the ruxolitinib arm [22].

The success of the REACH2 trial is also reflected in real-world data. In a retrospective analysis of 59 patients underwent treatment with ruxolitinib for SR-aGVHD, 36 individuals

(61.0%) demonstrated either a complete (CR) or partial response (PR) at the 28-day mark, while 31 patients (52.5%) achieved CR/PR by day 56. In summary, these findings from real-world clinical practice lend support to ruxolitinib as the standard of care for SR-aGVHD in clinical practice [23] (Table 1).

The REACH2 trial also identified some limitations of ruxolitinib use. A significant proportion of patients receiving ruxolitinib developed grade III or higher thrombocytopenia (27%) or anemia (22%). Ruxolitinib use in SR-aGVHD has been associated with an elevated risk of infections, as high as 68% in other studies [24]. The incidence of cytomegalovirus infection was 26% in the ruxolitinib cohort, which was modestly higher than the BAT cohort (21%). Subgroup analysis revealed that the overall response rate was lower among patients with grade III or IV aGVHD (54.9%), as well as those with involvement of liver or lower GI tract (57%).

These findings highlight a need to address ruxolitinib-refractoriness in this challenging population [25]. Mohty et al. have proposed the following criteria to define Ruxolitinib-refractory acute GVHD: (@) Progression of GVHD, either by increase in stage or grade compared with baseline, or new organ involvement, after 5–10 days of treatment with ruxolitinib; (@) Lack of improvement in GVHD symptoms compared with baseline after at least 2 weeks of treatment; or (@) Loss of response at any time after initial improvement [26]. Ruxolitinib-resistant disease is associated with particularly adverse outcomes, with limited options for effective treatment. In a retrospective review of 307 patients, ruxolitinib resistance was detected in 48 patients. The median survival for these patients was 21 days, with 36% ORR to subsequent therapy [27].

3. Novel investigational therapy for SR-aGVHD

Beyond ruxolitinib, there remains a lack of consensus on the next-line treatment for patients who are intolerant or refractory to ruxolitinib. Although the recent approval of ruxolitinib for SR-aGVHD has emerged as a beacon of hope, the management of SR-aGVHD remains an unmet clinical need. This comprehensively examines the landscape of emerging pharmacologic therapies poised to address the challenges posed by SR-aGVHD (Table 2).

3.1. Vedolizumab

Vedolizumab, an $\alpha 4\beta 7$ gastrointestinal integrin antagonist initially approved for inflammatory bowel disorders, has

Table 2. Novel treatments for SR-aGVHD evaluated in recent clinical studies.

	Best Response	Ongoing Clinical Trials	Phase
Vedolizumab [28–32]	35–100%	NCT06066840	Phase I
MSC [31–37]	35–82%	NCT03158896, NCT04629833, NCT02411084, NCT06321198, NCT04744116, NCT05333029 NCT04769895	Phase I – III
FMT [38–40]	38–100%		Phase III
CD3/CD7-IT [41–43]	60%		
Neihulizumab [44]	69%		
Begelomab [45,46]	75%		
Tocilizumab [23]	62.5–67%		

MSC: Mesenchymal stromal cells; FMT: Fecal microbiota transplantation.

shown efficacy in SR-aGVHD, particularly in GI involvement [47]. Vedolizumab emerged as a potential treatment option for SR-aGVHD after a proof-of-concept study demonstrated a response in all six of the patients who received vedolizumab [28]. In another study of 20 patients with steroid resistant lower GI aGVHD, day 28 ORR was reported as 35% (7 of 20 patients). Of note, vedolizumab demonstrated a considerable day 28 response of in 50% (6 of 12) ruxolitinib failures. Overall OS at 6 months was 35% [29]. Despite initial optimism, subsequent studies have yielded mixed results, with significant sepsis-related deaths reported in patients receiving vedolizumab. In a case series by Colcoff et al., 6 of 9 patients treated with vedolizumab died by day 28, and only one patient survived for longer than 2 months [30]. A phase 2a trial which examined vedolizumab in intestinal SR-aGVHD was terminated early as preliminary results failed to meet the primary efficacy endpoint [31]. The varied results highlight the need for further research to delineate its role in SR-aGVHD management, particularly in ruxolitinib-resistant or refractory disease; a randomized trial has been opened to investigate the efficacy of vedolizumab and basiliximab compared to basiliximab alone (NCT06066840) [32]*.

3.2. Mesenchymal stromal cell (MSC)

Mesenchymal stromal cells are cells derived from various tissue sources such as bone marrow, placenta, and adipose tissue, with the ability to differentiate into multiple cell types in vitro [33]. Through various paracrine, mitochondrial, and exosomal mechanisms, MSCs have demonstrated potential to induce immune tolerance and an anti-inflammatory response through regulating T lymphocyte activation, as well as antigen presentation, antibody production, and NK-cell cytotoxicity [34]. Furthermore, MSCs are believed to elicit minimal immune responses from the host due to their lack of expression of HLA class II and costimulatory molecules [35].

The initial report on utilizing MSCs for GVHD treatment was documented by the Karolinska Institute in Sweden. In this proof-of-principle report, haploidentical mesenchymal stem cells were infused to a patient experiencing severe treatment-resistant grade IV aGVHD, resulting in notable amelioration of gut and liver aGVHD symptoms [36]. These encouraging results led to several phase I/II studies reporting responses to MSCs in adults and children with SR-GVHD [37,38,48].

Unfortunately, a multicenter, randomized study which evaluated the efficacy of remestemcel-L, an ex vivo cultured human MSC product, alongside second-line therapy failed to demonstrate a difference in response rate at day 28 compared to placebo (35% versus 30%; $p=0.42$) [39]. In contrast, in another randomized phase 3 trial evaluating the MSCs in addition to basiliximab and calcineurin inhibitor therapy, the addition of MSC treatment modestly improved OR at day 28 (82.8% vs. 70.7%; $p=0.043$) [40]. These inconsistencies in reported outcomes have hampered the widespread adoption of MSCs in SR-aGVHD, although work is underway to analyze the contradictory results reported in these trials. The heterogeneity of donor source, culture methods, and expansion levels are considered to contribute to the variable efficacy of MSC [49].

MSC is currently not approved by the FDA for use in pediatric or adult SR-aGVHD. Biomarkers are being investigated to identify responders to MSC therapy [41]. As of March 2024, numerous trials are underway to investigate the use of MSC alone (NCT03158896, NCT04629833, NCT02411084, NCT06321198), or in conjunction with ruxolitinib (NCT04744116) or ECP (NCT05333029).

3.3. Fecal microbiota transplantation (FMT)

Intestinal dysbiosis is implicated in the pathophysiology of aGVHD, and the diversity of the intestinal flora at the time of engraftment is an independent predictor of mortality in allo-HSCT recipients. Fecal microbiota transplantation (FMT) from healthy donors has demonstrated potential in alleviating SR-aGVHD through restoring gastrointestinal microbiota diversity, enhancing resistance to potential pathogens, and modulating donor T cell function.

In a pilot study to assess the safety of FMT, four patients with steroid-resistant ($n=3$) or steroid-dependent ($n=1$) GI aGVHD were administered FMT. No severe adverse events associated with FMT were detected [42]. All patients exhibited a positive response to FMT, with three achieving complete responses and one experiencing a partial response. The temporal changes in microbiota dynamics appeared to correlate with the gastrointestinal condition of the patients, and there was also an observed increase in peripheral effector regulatory T cells during the response to FMT [42]. In a larger single-arm study of pooled allogeneic fecal microbiota MaaT013 for treatment of grade III-IV, steroid refractory GI-aGVHD, the day 28

ORR for GI aGVHD was 38% [43]. Of the 24-patient cohort, five cases of infectious complications, including three instances of sepsis, were reported. The study reported that the infections could not be excluded as a complication of FMT, although sequencing analyses of the identified infectious strains were not identified from the MaaT013 product. Furthermore, GI aGVHD is associated with an elevated baseline risk of bacteremia [44]. Currently, a phase III trial is investigating the role of FMT as salvage therapy in steroid and JAK inhibitors-resistant GI-aGVHD patients (NCT04769895).

3.4. CD3/CD7 blockade

Another approach to suppression of alloreactive donor T lymphocytes in GVHD is through the development of anti-T cell immunotoxins. T-Guard is comprised of anti-CD3 and anti-CD7 antibodies conjugated to immunotoxin ricin A, which reduces T lymphocyte activation and induces apoptosis of activated T cells [45]. In phase 1/2 trials, T-Guard has demonstrated significant complete response (CR) rates of 50% (10 of 20 patients) and overall response rates of 60% (12 of 20 patients) at day 28 among individuals with severe steroid-refractory acute graft-versus-host disease (SR-aGVHD) [46]. These positive results led to the inception of a single-arm phase III trial of T-Guard in SR-aGVHD (BMT CTN 1802).

Unfortunately, the BMT CTN 1802 trial was terminated after encountering three fatalities before study day 30. In a retrospective analysis of all 35 T-guard treated patients including those from prior trials, 30-day mortality was 26% (9 of 35 patients), with 4 fatalities attributed to refractory aGVHD and 5 to infection [50].

Consequently, with regulatory support from the FDA and the EMA, BMT CTN 2002, a pivotal phase III multicenter trial was pursued. This trial randomly assigned 12 patients suffering from severe SR-aGVHD to receive either T-Guard or ruxolitinib. BMT CTN 2002 implemented more stringent exclusion criteria compared to its predecessor, BMT CTN 1802, focusing on factors identified as potential contributors to early mortality, such as a body mass index (BMI) ≥ 35 , prior exposure to checkpoint inhibitors, or the presence of thrombotic microangiopathy.

Four of 7 patients in the T-Guard treatment group succumbed before day 60, in contrast to no deaths in the ruxolitinib group at day 60. The Data and Safety Monitoring Board (DSMB) recommended trial termination due to the elevated day 60 mortality rate (57%, [95% confidence interval (CI), 18–90%]) observed in subjects with SR-aGVHD treated with T-Guard [51].

3.5. Neihulizumab

Neihulizumab, targeting T cell apoptosis via P-selectin glycoprotein ligand-1 (also known as CD162), represents a promising avenue for SR-aGVHD treatment. Early-phase trials have demonstrated encouraging response rates, warranting continued investigation in larger studies. Preliminary results of 37 patients with SR-aGVHD or TR-aGVHD demonstrated Day 28 ORR of 69% and 50% in the two studied cohorts,

demonstrating a promising response [52]. There are no announced studies for neihulizumab, however we remain optimistic that this drug can be examined further for use in SR-aGVHD.

3.6. Begelomab

Begelomab, a monoclonal antibody against CD16 on T lymphocytes, impairs T cell migration and activation of cytokines. Begelomab has been investigated for use in several immune-mediated diseases, including GVHD [53]. In a trial comparing 28 patients with SR-GVHD to 82 matched controls, the day 28 ORR was 75% in the begelomab arm, compared to 41% in the control arm. The study did not identify any adverse events attributable to begelomab and reported a trend toward improved overall survival at 1 year (50% vs 31%; $p = 0.06$) [54]. In spite of these positive findings, NCT02411084, a phase 3 randomized trial comparing the efficacy of begelomab to conventional therapy, was terminated due to insufficient accrual rate.

3.7. Tocilizumab

Interleukin (IL)-6 is a cytokine with critical roles in numerous immune-mediated processes, including B- and T-cell differentiation and recruitment of T lymphocytes [55]. Mice models with GVH were associated with elevated expression of IL-6 and IL-6 receptor [56]. These findings have evoked interest in using tocilizumab, a humanized anti-IL-6 receptor monoclonal antibody, for the treatment of aGVHD.

In a retrospective analysis, Ganetsky et al. assessed the efficacy of tocilizumab in treating SR biopsy-proven lower GI aGVHD in 16 consecutive adult alloHCT recipients [57]. Tocilizumab at a dose of 8 mg/kg was administered every 2 weeks until patients achieved complete response, defined as the resolution of all GI GVHD manifestations, or until progression or initiation of alternative therapy occurred. Out of the 16 patients, 10 (62.5%; 95% CI, 0.39–82) achieved a complete response at day 28. In another study by Drobyski et al., tocilizumab was administered to eight patients with SR-aGVHD ($n = 6$) or SR-cGVHD ($n = 2$) every 3 to 4 weeks. Four (67%) of the six experienced either partial or complete response within 56 days of treatment initiation [58]. A recent study interrogating tocilizumab in combination with other agents such as itacitinib, a Jak1 inhibitor, has been terminated due to slow enrollment (NCT04070781).

3.8. Other therapies

Other monoclonal antibodies targeting T cell migration have also been investigated for use in SR-aGVHD. For instance, natalizumab, a humanized monoclonal antibody which inhibits T-cell trafficking to the GI tract through $\alpha 4\beta 7$ integrin, has been interrogated and has not shown improvement in outcomes in new high-risk aGVHD patients [59].

Basiliximab, a monoclonal antibody directed against the alpha chain of the interleukin-2 (IL-2) receptor, is proposed to improve aGVHD by limiting T-lymphocyte activation [60]. Basiliximab has demonstrated ORRs between 82.5% and 92%

[60–62] in several prospective single-arm trials. However, in a retrospective study comparing basiliximab to ruxolitinib, the day 28 ORR was higher in the ruxolitinib cohort compared to the basiliximab cohort (72.8% vs 54.2%). Furthermore, ruxolitinib was associated with a lower incidence of cGVHD and non-relapse mortality. There were also significant infection concerns raised in several of the studies examining basiliximab; in a study of 53 patients, 29 episodes of viral reactivations, 25 bacterial infections, and 11 probable fungal infections were reported [61].

Brentuximab is an antibody-drug conjugate targeting CD30, a cell surface marker on activated T cells. Therefore, brentuximab has been evaluated for use not only in lymphoproliferative disease but also in a phase I trial consisting of 34 patients with SR-aGVHD. Day 28 ORR was reported as 38.2%, with 5 patients achieving CR (14.7%) by day 28 and 7 additional patients achieving CR by day 56. Two patients experienced neutropenia leading to treatment-associated mortality, and therefore the dosing schedule was revised. Other grade 4 toxicities included thrombocytopenia, anemia, and sepsis. No peripheral neuropathy was reported [63]. Currently, there are no active trials investigating brentuximab for treatment of SR-aGVHD.

Alpha-1-antitrypsin (AAT), a serine protease inhibitor with anti-inflammatory effects, has been proposed to treat SR-aGVHD through increasing regulatory T cell (Treg) populations. AAT has demonstrated promising results in a single-arm, multi-center trial of 40 participants, 70% of whom had Grade 3 or higher SR-aGVHD [64]. By day 28, the overall response rate (ORR) was 65%, with 35% of patients achieving a complete response (CR). The day 28 response rate for gastrointestinal aGVHD was notable at 64%, while liver aGVHD patients had an ORR of 57%. AAT was well tolerated, with no Grade 3 or higher adverse events linked to the drug. The safety profile and promising results in gastrointestinal aGVHD have led to randomized trials investigating its role in treatment combined with steroid therapy (BMT-CTN 1705) and in prophylaxis of high-risk individuals.

Interleukin-22 (IL-22) is a tissue-protective cytokine which has been associated with tissue regeneration and mucosal healing [65]. Ponce et al. conducted a multicenter, single-arm phase 2 trial to assess the efficacy of a novel recombinant human IL-22, combined with systemic corticosteroids, for the treatment of newly diagnosed lower GI aGVHD. Nineteen of the 27 patients (70%) achieved a treatment response by day 28, with 13 patients (48%) achieving a CR. The most common adverse events were cytopenias and electrolyte abnormalities, with no dose-limiting toxicities observed [66]. Although no current studies are examining its role in steroid-resistant disease, targeting IL-22-mediated mechanisms may hold promise in future studies.

Pentostatin is a nucleoside analog that is a potent and irreversible inhibitor of adenosine deaminase. Pentostatin causes dampening of several T-cell functions, including interleukin-2 (IL-2) production as well as antibody and non-antibody dependent cytotoxicity. A phase I dose escalation study investigating pentostatin in 23 patients with SR-aGVHD demonstrated complete responses in 14 (63%) patients and partial responses in 3 (13%) patients after 1 cycle (28 days). Late infections at the 2 mg/m²/d dose level were identified as

dose-limiting toxicities. Based on these results, a phase II study was planned, however terminated due to CMV viremia and excessive pentostatin levels on AUC [67].

Promoting intestinal regeneration remains a promising new strategy for treating GI aGVHD. GLP-2-based treatment has demonstrated restoration of intestinal homeostasis in mouse models of GVHD [68]. Apraglutide, a potent and selective next-generation long-acting synthetic GLP-2 analogue, is currently being evaluated for the treatment of GI aGVHD in the phase 2 STARGAZE trial (ClinicalTrials.gov Identifier: NCT05415410). A recent phase II trial examined the concurrent use of human chorionic gonadotropin, which is essential to immune tolerance in pregnancy, and epithelial growth factor, which promotes mucosal repair, in 22 high-risk aGVHD patients. ORR at day 28 was 68%, with 57% of patients achieving a complete response [69]. Given impressive responses for GI aGVHD, further studies in the steroid-refractory population may be warranted.

4. Conclusion

SR-aGVHD remains a formidable complication post-HCT, and the development of innovative therapeutic approaches remains a clinical priority. While the advent of ruxolitinib and novel approaches have improved outcomes, challenges persist, particularly in high-risk patient populations and in GI-predominant disease.

Despite positive results in earlier phase studies, many agents fail to replicate a robust response in randomized, multi-center trials. Harris and colleagues propose that variability in assessing GVHD response may be contributing to the lack of reproducibility, calling for a more stringent and standardized GVHD grading [70]. There is also interest in developing a personalized, biomarker-driven approach to SR-aGVHD treatment. The Mount Sinai Acute GVHD International Consortium (MAGIC) has been successful in validating biomarkers ST2 and REG3 α to predict NRM and OS in steroid-resistant GVHD patients [71].

The pursuit of novel agents and combination therapies holds promise for further enhancing SR-aGVHD management. The results of current trials investigating FMT (NCT04769895), MSC (NCT03158896, NCT03158896, NCT04629833, NCT02411084, and NCT06321198) are eagerly awaited.

5. Expert opinion

Steroid-refractory acute GVHD remains a critical complication after allogeneic transplant. The clinical efficacy and regulatory approval of ruxolitinib phosphate have made it the standard for SR-aGVHD, however long-term experience has shown that nearly half of patients developing SR-aGVHD on ruxolitinib still perish due to either GVHD or complications that arise during GVHD treatment, namely infection and relapse. Therefore, clinical development of novel therapies for SR-aGVHD continues to remain a significant unmet need.

In order to improve outcomes for SR-aGVHD, investigation into multiple avenues are likely needed to discover the next breakthroughs. One approach warranting investigation involves the identification of patients at risk for the development of SR-aGVHD. Biomarker studies at acute GVHD diagnosis, namely

measurement of ST2 and REG3a levels, are a highly sensitive tool to distinguish patients with high-risk aGVHD, i.e. patients likely to have persistent aGVHD after initial corticosteroid therapy. The importance of early identification of high-risk patients is that a population can be identified that may benefit from additional upfront therapy to corticosteroids while also sparing lower risk GVHD patients of additional toxicities. Initial corticosteroid combination trials for aGVHD, while initially demonstrating augmented efficacy, largely failed due to added toxicities, hence highlighting the importance of selection. This approach has been put into practice in a BMT-CTN sponsored trial adding alpha-1 antitrypsin to corticosteroids in high-risk aGVHD. This study has completed enrollment, and at the time of this manuscript, read out is pending. At a minimum, this trial represents proof of principle that future studies can follow. Further, it is now understood that among patients with SR-aGVHD, responses to ruxolitinib are more limited among higher grade (III-IV) patients and patients with severe lower GI and liver involvement. Novel combinations at the time of SR-aGVHD diagnosis could be explored in these populations, potentially with agents known to have GI and/or liver targeting effects. A single-center study combining ruxolitinib and chorionic gonadotropin demonstrated promising results and represents an example of efforts that could be taken. Other therapies being combined with ruxolitinib in the SR-aGVHD setting include ECP. A single-center study of combining ruxolitinib with ECP in 18 patients with severe GI SR-aGVHD demonstrated promising results, with CR in 44% and PR in 11%, as well as an estimated 2-year overall survival of 56%. No severe cytopenia (grade IV) occurred, although 3 of 18 patients experienced grade III cytopenia [72]. Finally, trials addressing treatment-refractory acute GVHD, that is steroid and ruxolitinib refractory acute GVHD, represent an important area of need. These patients are inevitable and have a dismal prognosis of only one month. As novel immunotherapies are developed addressing treatment-refractory inflammatory bowel disorders or autoimmune skin disorders, consideration, and investigation of these agents for treatment-refractory acute GVHD should be a priority. Neihulizumab is an example of one such agent, with modest efficacy in early clinical trial in this setting. Inhibitors of sphingolipid metabolism, including ozanimod, may hold promise. Preclinical studies indicate that inhibition of sphingosine-1-phosphate (S1P) signaling has the potential to attenuate acute GVHD through modulation of T-cell survival, function, and migration [73]. Finally, inhibitors of Bromodomain and Extra Terminal (BET), which are clinically available, may be novel agents to explore to mitigate acute GVHD, as preclinical studies demonstrate that targeting BET appears to uniquely dampen dendritic cell and T-cell function, mitigating acute GVHD [74].

In summary, novel combinations, either when high-risk aGVHD or steroid-refractory aGVHD represent a pathway forward that should improve outcomes. Furthermore, investigation into novel agents, potentially first in the treatment-refractory setting, then subsequently in earlier lines, may provide a pipeline and path forward to additionally incrementally improve outcomes.

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